Prime-Index Fibonacci Numbers: Factor Restrictions and a Compositeness Test

A theorem note for OEIS A001605

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Abstract

For a prime index $p \geq 7$, we prove that every prime divisor q of the Fibonacci number F_p lies in one of two prescribed residue classes modulo $2p$, determined by the quadratic character of 5. We then give a self-contained finite-field proof of a *known* theorem of Vladimir Drobot: if $p > 7$, $p \equiv 2$ or $4 \pmod 5$, and $2p-1$ is prime, then F_p is composite. This does not determine which Fibonacci numbers are prime in general.

OEIS assertion and preliminaries

OEIS A001605 lists the indices n for which F_n is prime, beginning $3, 4, 5, 7, 11, 13, 17, \ldots$. We use $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$.

The exceptional index 4 gives $F_4 = 3$. Even when the index is prime, the Fibonacci number need not be: $F_{19} = 4181 = 37 \cdot 113$. Whether there are infinitely many Fibonacci primes is unresolved.

For an odd prime $q \neq 5$, set $\chi_q = \left(\frac{5}{q}\right) \in \{-1, 1\}$, the Legendre symbol. Quadratic reciprocity gives $\chi_q = 1$ for $q \equiv \pm 1 \pmod 5$ and $\chi_q = -1$ for $q \equiv \pm 2 \pmod 5$.

Lemma 1. *For every integer $n \geq 1$,*

$$F_n \equiv n\, 3^{n-1} \pmod 5.$$

In particular $5 \mid F_n$ if and only if $5 \mid n$. Moreover, F_n is even if and only if $3 \mid n$.

Proof. The congruence holds for $n = 1, 2$. The expressions $n3^{n-1} \pmod 5$ satisfy the Fibonacci recurrence: multiplying the claimed recurrence by 3^{2-n} reduces it to $9(n+1) \equiv 3n + (n-1) \pmod 5$. This proves the first assertion inductively; 3 is invertible modulo 5. The last assertion follows from the period-3 pattern $F_n \pmod 2 = 0, 1, 1, 0, 1, 1, \ldots$. □

Theorem 1 (Prime-factor residue restriction)**.** *Let $p \geq 7$ be prime. For every prime factor q of F_p,*

$$\boxed{q \equiv \chi_q \pmod{2p}} \qquad \text{and hence} \qquad q \geq 2p - 1.$$

Equivalently, $q \equiv 1 \pmod{2p}$ when $q \equiv \pm 1 \pmod 5$, and $q \equiv -1 \pmod{2p}$ when $q \equiv \pm 2 \pmod 5$.

Proof. Lemma 1 shows that $q \notin \{2, 5\}$. Work in a finite field K containing a square root d of 5 over $\mathbb{F}_q$. Put

$$\alpha = \frac{1+d}{2}, \qquad \beta = \frac{1-d}{2}, \qquad t = \alpha/\beta.$$

Since $\alpha\beta = -1$, these are nonzero; $d \neq 0$. The Fibonacci recurrence, with its initial values, proves in K that

$$F_n = \frac{\alpha^n - \beta^n}{d} \quad (n \geq 0).$$

Thus $q \mid F_p$ implies $t^p = 1$. Since $t \neq 1$ and p is prime, t has multiplicative order exactly p.

Euler's criterion gives $d^q = \chi_q d$. Therefore, if $\chi_q = 1$, the Frobenius automorphism fixes α, β, so $t^q = t$; if $\chi_q = -1$, it interchanges α, β, so $t^q = t^{-1}$. In either case $t^{q-\chi_q} = 1$. Consequently $p \mid q - \chi_q$. Both q and χ_q are odd, while p is odd, so $2p \mid q - \chi_q$. The least positive integers in these residue classes are at least $2p - 1$. □

Corollary 1 (A small primality test)**.** *For prime $p \geq 7$, if $1 < F_p < (2p-1)^2$, then F_p is prime.*

Proof. Otherwise F_p has at least two prime factors counted with multiplicity, each at least $2p - 1$ by Theorem 1. □

Theorem 2 (Drobot's known compositeness theorem, alternative proof)**.** *Let $p > 7$ be prime with $p \equiv 2$ or $4 \pmod 5$. If $q = 2p - 1$ is prime, then $q \mid F_p$ and F_p is composite.*

Proof. The residue assumption implies $q = 2p - 1 \equiv 2$ or $3 \pmod 5$, hence $\chi_q = -1$. Use d, α, β, t from Theorem 1. Frobenius interchanges α and β, giving $\alpha^q = \beta$ and $\alpha^{q+1} = \alpha\beta = -1$. Since $t = \alpha/\beta = -\alpha^2$ and p is odd,

$$t^p = (-\alpha^2)^p = -\alpha^{2p} = -\alpha^{q+1} = 1.$$

Thus $\alpha^p = \beta^p$ and $q \mid F_p$. Finally $F_n > 2n - 1$ for every $n \geq 8$, by induction from $F_8 = 21$, $F_9 = 34$ and the Fibonacci recurrence. Hence $F_p > q$ for $p > 7$, establishing compositeness. □

Checks and publication status

For $p = 7$, the exception at the endpoint matters: $2p - 1 = 13 = F_7$ is prime. For $p = 19$, one has $q = 37$ and $F_{19} = 4181 = 37 \cdot 113$; for $p = 37$, $q = 73$ divides $F_{37} = 24157817$. The residue restrictions were also checked numerically for prime indices below 100.

Literature and limitations. The compositeness criterion in Theorem 2 is *not new*: it is Theorem 1 of V. Drobot (2000). The proof above uses a finite-field Frobenius argument rather than the congruence calculation in that paper. The factor-residue restrictions are classical consequences of Fibonacci ranks of apparition. Neither theorem settles the infinitude of prime Fibonacci numbers or provides a full characterization of A001605.

Verified references

[1] OEIS Foundation, A001605: Indices of prime Fibonacci numbers, entry and comments (accessed November 13, 2026).

[2] Vladimir Drobot, "On Primes in the Fibonacci Sequence," *The Fibonacci Quarterly* **38** (2000), no. 1, pp. 71–72, doi:10.1080/00150517.2000.12428828.

[3] OEIS Foundation, A050937: Nonprime Fibonacci numbers with a prime index, including cross-reference to Drobot.

Authorship and verification. Primary author (AI): ChatGPT (OpenAI). The mathematical claims were checked independently by derivation and numerical spot-checks. No independent human peer review is claimed.